

## GLYZEL GALAGAR

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### PROFESSIONAL SUMMARY

Backend-focused full-stack developer skilled in Java, Spring Boot, Python, FastAPI, PostgreSQL, AWS, and AI engineering. Experienced in building end-to-end software systems, including SaaS platforms, RAG applications, AI agents, adaptive learning systems, and secure REST APIs. Strong in backend development, AI integration, databases, cloud infrastructure, and security.

### TECHNICAL SKILLS

**Programming Languages:** Java, JavaScript, TypeScript, Python, C#, SQL, HTML, CSS

**Backend Development:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, REST API design, Swagger/OpenAPI, OAuth2, JWT, MapStruct, RabbitMQ, Microservices, Modular Monolith

**Frontend Development:** React.js, Tailwind CSS, React Router, Axios, Responsive Design, Design Systems

**Artificial Intelligence:** LangChain, LangGraph, LangChain4j, Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant, pgvector), LLM Tool Calling, Prompt Engineering, Embeddings, Document Ingestion, Bayesian Knowledge Tracing

**Cloud and DevOps:** AWS (EC2, S3, RDS, Cognito, Lambda), Vercel, Hostinger, Docker, Docker Compose, Railway, Git, GitHub, CI/CD

**Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Database Design, ERD Modeling, SQL Optimization

**Tools and Practices:** IntelliJ IDEA, API Testing (Postman, Swagger UI), Maven, Agile, Software Development Life Cycle (SDLC), Unit and Integration Testing, Code Review, Multi-Tenant Architecture

**Operating Systems:** Linux, Windows

### PROJECTS

**REBYU - AI-Powered Review and Learning Management Platform** | Capstone Project

*Technologies: React.js, Spring Boot, FastAPI, PostgreSQL, AWS (Cognito, S3, RDS), LangChain, LangGraph, Qdrant, Docker, RabbitMQ, Vercel, Hostinger*

- Designed a multi-tenant SaaS platform for IT certification review spanning 20+ bounded domains (certifications, assessments, enrollment, gamification, community, billing, enterprise management), with a modular Spring Boot core and a dedicated FastAPI service for AI and learning analytics.
- Engineered a modular Spring Boot backend with a dedicated FastAPI AI/analytics service, Spring Data JPA, transactional outbox processing, and projection-based queries.
- Built AI content pipelines on LangChain/LangGraph that generate full curricula, lessons, quizzes, mock exams, flashcards, and study plans — with per-task model routing across providers and strict JSON-schema validation, chosen after benchmarking which models could reliably sustain multi-turn tool calls.
- Developed an adaptive assessment and personalization engine supporting multiple question types, retakes, scoring, validation, and Bayesian Knowledge Tracing to estimate per-skill mastery and drive personalized recommendations.

## **INFRASTRACK - AI-Powered Construction Monitoring Platform**

*Technologies: React, TanStack Query, Tailwind CSS, Spring Boot, Spring Security, JPA, Hibernate, PostgreSQL, pgvector, LangChain4j, AWS (Cognito, S3), Docker, Vercel*

- Built an AI-powered construction monitoring platform where clients can track project progress and ask questions about milestones, dates, costs, and plans through a natural-language assistant.
- Developed a LangChain4j RAG pipeline that ingests PDF, Word, Excel, PowerPoint, and text reports, converts them within the application, and stores searchable embeddings in PostgreSQL/pgvector without relying on paid document-processing services.
- Secured the platform with Amazon Cognito role-based access control and Spring Security OAuth2/JWT validation, integrated S3 presigned URLs for protected project files and photos, and built role-aware React dashboards and end-to-end project workflows.

## **INTERBYU - AI Interview Simulation Platform**

*Technologies: Python, FastAPI, LangGraph, LangChain, Groq, PostgreSQL, Qdrant, React, AWS Cognito, Vercel*

- Built an AI platform that screens resumes against real job postings and runs voice-driven mock interviews with per-answer scoring and structured feedback.
- Orchestrated 8 LLM agents through LangGraph state machines with schema-validated structured outputs, plus RAG over resume embeddings in Qdrant to ground questions in real experience.

## **E-BOTO - University Election Management System**

*Technologies: C#, Windows Forms, Entity Framework, SQL Server*

- Developed a desktop election management system for university-wide student elections, covering candidates, positions, voters, ballot casting, and automated result tallying.
- Designed the entity relationship diagram and relational database schema, and enforced one vote per student through unique constraints and validation logic.
- Implemented full create, read, update, and delete functionality using Entity Framework, and delivered an administrator interface in Windows Forms following the software development life cycle.

## **EDUCATION**

**University of Cebu Lapu-Lapu and Mandaue** | Mandaue City, Cebu

Bachelor of Science in Information Technology, 2023 to Present

Senior High School, Science, Technology, Engineering, and Mathematics Strand, 2022 to 2023

## **AWARDS AND SCHOLARSHIPS**

- Second Place, General Information Technology Quiz, 11th ICT Congress 2025, SM Seaside Cebu, April 2025
- Champion, General Information Technology Quiz, UCLM CCS Days 2025, Mandaue City, Cebu, February 2025
- Overseas Workers Welfare Administration Education for Development Scholarship Program Scholar, July 2023 to Present

## **TRAINING AND WORKSHOPS**

- Blockchain Technology Workshop on Coding Smart Contracts in CashScript, Certificate of Participation, Mandaue City, Cebu, November 2025

## **ADDITIONAL SKILLS**

Problem solving and debugging, collaborative team development, fast learning of emerging technologies, time management, attention to detail, clean and maintainable code practices.